I'm going to read you the opening of my article, Escaping False Valleys.

She had stopped trusting the mountain about an hour ago.
After the storm, the trail had disappeared and the light dropped faster than expected. Her phone battery was nearly dead, not that it mattered much because she had no reception anyway. Somewhere beyond the thick scrub and the folds of dark red rock there was a town – a pub, a servo, a road, a bed, civilisation. And she knew one useful thing – the town sat in the deepest valley in the range.
So she did what felt obvious. She rode her mountain bike downhill. And for a while it felt like progress.
Until it didn't.
That was when she realised the problem was not the mountain. It was her method.
She was using a greedy algorithm.

My name is Saf Flatters, and I'm studying a Bachelor of Data Science.

And yes — my article is a piece of fiction.

I'm calling the genre Statistical Fiction. Or **Stat-fi**.

I wrote about a mountain-bike rider lost in a mountain range to explore some of the concepts I've been learning during my degree.

I love science fiction — or really, anything that mixes adventure and drama with the chance to learn something technical along the way.

And that's actually part of what led me to data science.

I started my degree a little later in life, and my interest in maths definitely didn't come from a lifelong love of equations.

It came from books like *The Three-Body Problem, The Martian* or anything by *Ted Chiang*. Stories like these put complex ideas into practical settings I could visualise before I needed to fully understand the maths.

They made me want to learn more.

So now, in my final semester, when I saw the Royal Statistical Society and *Significance Magazine* writing competition, I thought I'd give it a crack.

Could I use fiction to make someone else curious about maths?

Could I use fiction to TRICK a non-mathematical audience into reading about…

[jazz hands]

statistics!

There was one slight problem with this plan.

I had to understand the statistics well enough to write about it.

And I'd learnt long before university that explaining **why** something works is a pretty good test of whether you actually understand it.

I'm actually an electrician, and I'd spent years teaching apprentices *how* to do things. But the interesting part was always trying to explain **why we do things that way**.

The *"that's just how we do it" answer* is such a cop out.

Now at university, I find myself doing exactly the same thing.

I like standing at a whiteboard and trying to explain what I'm learning to anyone who'll listen.

I'll draw something, realise my explanation doesn't check out, rub it out, discuss it and try again.

And somewhere in that process, I would actually learn the concept properly.

So this article became another version of that test:

Could I understand the *how and why* well enough to explain it

— and then make it interesting enough that someone would actually want to read it?

I also decided to draw and paint all the plots for my article by hand.

I'd been inspired by some of the beautiful historical statistical graphics we learned about in Data Visualisation.

And then for THIS presentation, I decided I needed an excuse to draw some more — so I painted some of the story too.

Many of the following drawings may have been inspired by my own personal mountain biking experience. I love riding and I've had my fair share of experiences of being lost on a trail.

This story is about a rider looking for a town that sits in the lowest valley in the range. The problem is that the mountain range is full of other low points like gullies, basins and false valleys that feel like progress but it may not be the lowest part of the range.

At every decision point, she had greedily chosen the steepest immediate drop, assuming that what looked best locally would lead to the best outcome globally. Sometimes that works. Sometimes the landscape is simple and predictable, and downhill really does mean better. But this mountain range was messy. It was full of little bowls and basins, all lower than the ground around them but nowhere near the true lowest valley. If the mountain were a function, she was trying to minimise it using only local information, and the function was ug-ly! The town was the global minimum, and these gullies were just decoys.

She stood there with one hand on the bars and tried to think like a statistician instead of a panicked rider trying not to die in the bush.

Okay, I've made this ridiculous promise to hide statistics inside a mountain-bike story. So what statistics am I actually trying to hide?

These were three concepts that were going to save our poor rider from certain doom:

global optimisation

stochasticity

Bayesian reasoning

First up: **global optimisation.**

I wanted the reader to understand that when we build a model with multiple variables, we are often searching for the set of choices that gives us the lowest error.

Here is another except from my story:

Working out the best answer from incomplete information becomes a search problem. Suppose an outcome is being estimated from several different variables. Each variable is given a weight, which determines how much influence it has on the result. But there is no obvious way to choose those weights correctly from the start. Different choices of weights produce different answers, and some are more wrong than others. The task is to find the set of weights that makes that error as small as possible.

With one variable to adjust, the error forms a simple curve, and the lowest point on that curve represents the answer closest to the truth

With two variables, the error forms a surface – something more like a landscape with slopes and valleys. Each position represents a different combination of weights, and the height of the surface shows how wrong those choices are.

But real problems rarely stop at two variables. Each additional variable adds another direction the model can move, and the landscape stretches into more dimensions than can be seen. It still exists, but it can no longer be fully drawn or pictured. It simply has to be accepted that it is there, and that every decision is a step through a space that cannot be fully visualised, guided only by whether things are getting better or worse. The problem then becomes a search for the position in this landscape that makes the error as small as possible – the global minimum.

My favourite part about writing this section was trying to visualise a multivariable problem. I called my mum and said "I'm going to try and explain multidimensional spaces to you". My poor mum.

But that conversation helped me make the metaphor work so I could write about it. In the story the rider is moving through a physical landscape. 2 dimensions are longitude and latitude. The 3rd is elevation but in the statistical metaphor, elevation becomes error. Where lower the valley, the lower the error. And the lowest valley, is town, the ideal combination of variables.

But the rider is inside the landscape, not looking down on it. She can not see the whole range at once. This is what make it useful as a metaphor for higher dimensional problems. In real modelling problems, we cannot fully visualise all the dimensions we are moving through. But we still need to search that space for the lowest error.

So what kind of search strategy should we use?

Her downhill-only strategy had failed because it kept exploiting the best local option. What she needed now was a strategy that allowed for exploration – a way to avoid getting trapped in the same kind of dead-end low point over and over again. That meant doing something that felt completely wrong – sometimes choosing a trail that went uphill. But not every time. She needed a way to be random on purpose.

Randomness seems to turn up everywhere in statistics, doesn't it? In random sampling, where it helps us make inferences from just part of a population; in simulation, where it helps us model variation; and in machine learning and optimisation, where stochastic methods can stop a search becoming too rigid too early.

To a non-statistical audience, deliberately adding randomness to find an answer sounds like a terrible strategy.

So, obviously, I HAD to write about how adding randomness can make a model more robust.

This is where the story connects to an algorithm called simulated annealing. Early in the rider's search, the rider uses randomness to move away from "best looking options" to be able to escape false valleys. That freedom to explore is gradually dialled back as her decision method becomes more selective as she gets closer to the final drop.

The final concept I wanted to try writing about was Bayesian reasoning. This one matters to me personally, because reading about Bayesian got my attention waaay before I started formally learning about it.

I've spent about 15 years troubleshooting electrical and mechanical faults, and fault finding can feel chaotic.

But most of us tradies probably don't realise that there's something quite Bayesian about the way we do it. We start with an idea of what we think is most likely, based on our previous experience. Then we test something and collect new evidence, update what we think is going on, and repeat.

Each new piece of evidence shifts the likelihood of the possible causes, until eventually we narrow it down the fault cause.

Could I apply that same decision pattern to being lost in a mountain range?

She started with one prior belief: the town sat in the deepest valley in the range. But knowing that was not the same as knowing how to get there.
So as she moved, she updated her thinking with every new observation. A narrow basin with no outflow made that path seem less likely. A slope that joined a more substantial creek made it seem more likely. The faintest suggestion of cattle fencing shifted the odds again. This is Bayesian reasoning in the wild: start with a belief, gather evidence, update belief, act and repeat.
She stopped thinking, "Is this the right path?" and started thinking, "Given what I know, which path is most probable – and when should I gamble against my current model?"

To me, being aware of Bayesian reasoning when under pressure can bring structure to the chaos.

Maybe my next STAT-FI project can be about linking data science with decisions made by technicians?

The mountain no longer felt like her nemesis, just an incomplete dataset.
The best route down was not the steepest one beneath her feet. It was the one she could only

find by first escaping the local geometry.
The mountain had not required heroism. It had required statistical thinking:

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the willingness to stop treating the nearest, easiest answer as the best one.

To accept uncertainty.

To search through a landscape she could not fully see.

To update beliefs.

And to allow a little randomness into the search.

We have been using stories to share knowledge for as long as humans have existed. I hope you enjoy mine, and I hope it inspires others to be as curious about mathematics and data as I am. Oh! And any Australian slang you didn't understand – I'm happy to have a chin wag with you after!

Thank you.
